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Write here your dedication

Acknowledgment

Write here the acknowledgements and grants, if any

Resumo

Write here your abstract in Portuguese

Abstract

Write here your abstract in English

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List of Acronyms

CHAPTER 1

Introduction

1.1. Context

The global pandemic triggered in 2019 by the **COV!** (**COV!**) and declared by the World Health Organization on March 11, 2020 [?], has exposed inherent weaknesses in health systems worldwide.

This dissertation aims to contribute to the development of a mobile health application enhanced by an Artificial Intelligence (AI) in a project coordinated by the Center for Research in Science, Technology and Architecture (ISTAR-Iscte), under grant number XYZ.

1.2. Motivation

This research is motivated by altruistic aims.

1.3. Problem

The central challenge lies in the scarcity of advanced mHealth applications capable of efficiently collecting health data, performing continuous monitoring, accurate diagnosis, and offering personalized health services, all while guaranteeing the privacy and security of user data.

1.4. Objectives and Research Questions

Considering the challenges presented in the research,

RQ1: ...

RQ2: ...

1.5. Methodology

Our methodology is based on Design Science Research (DSR) to attain its goals and answer the research questions.

CHAPTER 2

Literature Review

This section will look at Federated Learning techniques and data processing in mobile health (mHealth), exploring the application of machine learning methods and real-time signal processing for analyzing medical data. Techniques aimed at the early detection of anomalous patterns, such as changes in heart rhythms, which can indicate potential health problems in users in mHealth environments, will be discussed.

2.1. Systematic Literature Review

Following the **PRISMA!** (PRISMA!) methodology¹, the search for papers related to the dissertation had to be carried out in several repositories. The repositories used were the IEEE Xplore Digital Library², Web of Science database³ and Scopus database⁴.

2.2. Discussion of Related Work

In recent years, there has been a significant increase in the number of articles exploring how

2.2.1. Federated Learning in XYZ

The literature review highlights

2.2.2. Federated Learning in Cardiovascular and Neurological Health

An analysis of studies focusing on the use of Federated Learning in cardiovascular and neurological health reveals significant advances.

2.2.3. Federated Learning and COVID-19

The literature has explored various strategies to address the COVID-19 pandemic through Federated Learning, resulting in a wide variety of articles dedicated to this topic.

2.2.4. Summary

Although there are several mobile healthcare applications that employ .

¹PRISMA: <http://www.prisma-statement.org/>

²IEEE Xplore: <https://ieeexplore.ieee.org/Xplore/home.jsp>

³Web of Science: <https://www.webofscience.com/wos/woscc/basic-search>

⁴Scopus: <https://www.scopus.com/search/form.uri?display=basic#basic>

CHAPTER 3

Methodology

...

CHAPTER 4

Results

...

CHAPTER 5

Added Things: some extra things that might be helpful

This chapter isn't in the template provided by ISCTE and contains some extra things that might be helpful.

5.1. Glossary

This section explains how to use a package that handles the abbreviations. Near line 107 there's the following code:

```
\DeclareAcronym{wip}{  
    short=WIP,  
    long=work in progress,  
}
```

Be WIP an acronym for "work in progress" and "wip", the first time `\ac{wip}` is written, the package writes **wip!** (**wip!**). The next time it's used, it only writes **wip!**.

Pay attention to acronyms with very long meanings as the package does not handle them well.

5.2. Lists

An example of a bullet list

- Potatoes
- Lettuce

An example of an ordered list

- (1) A
- (2) B
- (3) C

5.3. Bibliography

To reference the bibliography, write the bib of the reference in the references.bib file then use `\cite{}`. Example: [?]. The bibliography is already getting added and numbered according to when they are referenced.

5.4. Tables

Check the code on this section to see how to make a table. Both tables and images might not appear where you expect them to appear to save up space.

5.5. Images

Here are two examples of how to place images in the document.

A	B	C
1	2	3
a	b	c
0	99	

TABLE 5.1. Caption

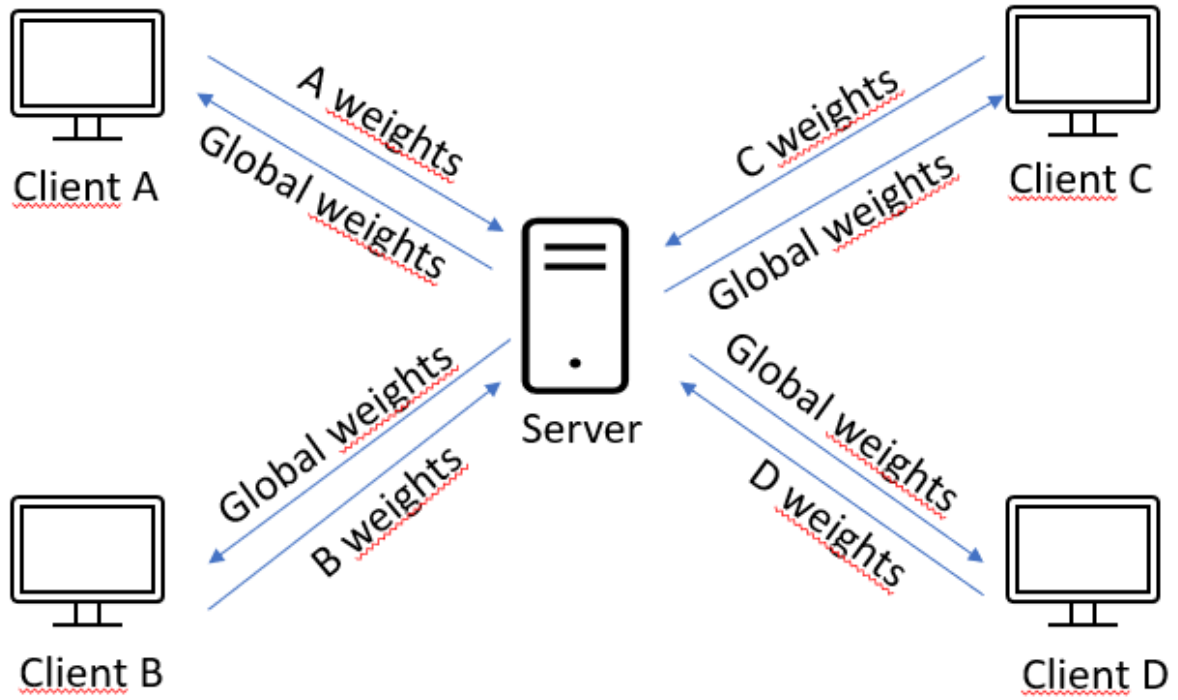


FIGURE 51. An example of a picture

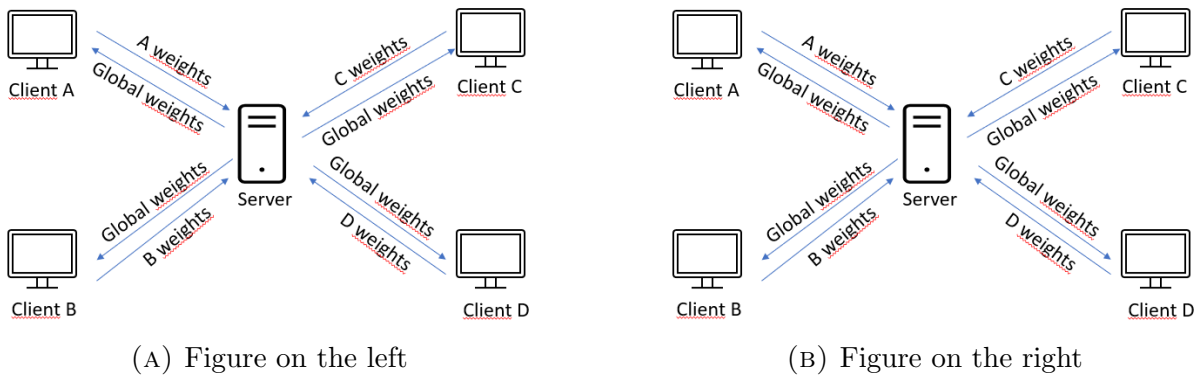


FIGURE 52. I reused the same picture

CHAPTER 6

Conclusions

APPENDIX A

An appendix